

Haiyang Zhang

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PROFESSIONAL SUMMARY

Senior Frontend Engineer with 8 years of experience building complex enterprise platforms, workflow-driven tools, and cross-platform healthcare applications. Strong at turning system complexity into usable, scalable, and high-performance product experiences through front-end architecture, platform standardization, performance optimization, and healthcare data integration.

CORE SKILLS

- Languages: HTML, CSS, JavaScript, TypeScript
- Frontend: Vue.js, React, React Native, Expo, jQuery, Backbone.js
- State Management: Vuex, Pinia, Zustand
- Testing: Jest, Cypress
- Tooling: Webpack, Vite
- Architecture: Front-End Architecture, Component Design, Performance Optimization, Workflow Systems
- Backend & Integration: Node.js, Express.js
- Healthcare & Web3: FHIR, EHR Integration, Healthcare Data Processing, ethers.js, Blockchain Application Development

PROFESSIONAL EXPERIENCE

Novo Vivo

Software Developer, Applications / Software Engineer Sep 2018 - Present

Distributed Privacy Computing Data Service Platform

2018 - 2021

- Led front-end development for a large-scale privacy computing platform spanning model management, inference testing, online and offline workflows, access control, service monitoring, and operational dashboards across multiple interconnected subsystems.
- Built complex interfaces for model listing, model detail, inference configuration, execution history, and result inspection, improving end-to-end workflow usability and reducing manual operational steps for users.
- Collaborated closely with backend and algorithm teams to define API contracts and integrate asynchronous task flows for model deployment, runtime tracking, and result delivery, helping streamline cross-system interactions.
- Optimized large data tables, search and filter workflows, and detail views, reducing response and interaction latency by an estimated 30%+ in core management screens.
- Improved rendering performance and state handling for complex task and monitoring pages, cutting first-screen load time for key views by approximately 25% to 35%.
- Implemented permission-aware UI controls and operational monitoring capabilities while coordinating front-end delivery across parallel subsystems and mentoring up to 4 engineers.

Standardized Privacy Computing Platform

2021 - Present

- Helped evolve a first-generation project into a standardized privacy computing platform by redesigning front-end architecture around reusable capabilities, lower coupling, and faster delivery for follow-on customized projects.
- Abstracted common capabilities across data resource management, federated modeling, joint statistics, privacy-preserving query, approval workflows, and system management instead of continuing one-off module implementations.
- Built reusable business components and shared interaction patterns that reduced duplicated front-end implementation by an estimated 30% to 40% across similar modules and later project variants.
- Reframed workflow orchestration from page-level implementations to more granular workflow and computation components, enabling more flexible drag-and-drop composition for multiple process types.
- Optimized workflow-canvas and task orchestration pages by reducing repeated re-rendering in high-interaction scenarios and improving first-screen rendering and responsiveness by an estimated 35%+ after architectural optimization.
- Helped turn a project-oriented solution into a reusable product foundation adopted by multiple customized client implementations.

Lavita Health App

2023 - Present

- Developed a cross-platform healthcare application for iOS and Android using Expo, delivering AI-driven health guidance in a privacy-focused product environment and supporting 100,000+ downloads across both platforms.
- Built core ingestion and processing flows for personal health records, enabling users to import medical data from major EHR systems including Epic and Cerner.
- Implemented FHIR-based healthcare data transformation and normalization logic, converting standards-compliant medical records into product-ready data for downstream AI workflows.
- Added fallback handling across the import flow so issues in individual processing steps would not break later API requests and block downstream data retrieval.
- Integrated Google NotebookLM to generate healthcare blog and podcast content, and built Web3 token flows on Theta's subchain including registration, balance retrieval, and token transfer.
- Supported privacy-first health-data flows using a hybrid encryption approach, combining asymmetric key exchange with symmetric encryption for larger payloads before downstream processing.

EDUCATION

Syracuse University

M.S.E. in Computer and Information Science, 2018

Beijing University of Technology

B.S.E. in Internet of Things, 2016